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Beyond One-Size-Fits-All: How Adaptability and Autonomy Shape Responsible AI Governance

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Abstract

Artificial intelligence (AI) governance is a critical safeguard against risk and ethical harm. However, organizations need clearer guidance on how oversight should vary across an AI portfolio. Building on characteristics recognized in scholarly work, we organize governance around two dimensions: adaptability, the degree to which a system's behavior changes through learning after deployment; and autonomy, the degree to which it selects or executes consequential action without contemporaneous human direction. Adaptability can make earlier validation and documentation progressively less representative of the system in use, while autonomy enables consequential actions without immediate human discretion and raises questions about intervention and accountability. We develop a two-dimensional model that distinguishes these governance demands and identifies when they become recursively compounded: system action shapes what is subsequently observed, those observations inform post-deployment learning, and the resulting adaptation changes subsequent actions. In these situations, the governance of action and learning can no longer proceed independently. We extend this analysis to portfolio interdependence, showing why organizations need to trace shared dependencies, assign responsibility, and coordinate review when a change or failure in a shared component affects multiple AI applications. Our model presents responsible AI governance as an organizational capacity for matching oversight to changing systems, delegated action, and interdependencies across an AI portfolio.

Keywords: AI governance; organizational AI; adaptability; autonomy; AI agents; algorithmic accountability; responsible AI

1 INTRODUCTION

Organizations now deploy AI systems for hiring and selection, consumer credit, and clinical diagnosis (Lebovitz et al., 2022; Sargeant, 2023; Schlicht & Råker, 2024). These systems inform managerial decisions and can carry out consequential tasks, such as completing transactions or coordinating work (Ezell et al., 2025; Hillebrand et al., 2025). When they fail or behave unexpectedly, the consequences extend well beyond internal compliance. Discriminatory outcomes can restrict access to employment, while opaque decisions can make it harder for applicants to understand a rejection or exercise their right to complain (Hunkenschroer & Kriebitz, 2023). AI governance provides an organizational response to these ethical and societal stakes by connecting responsibility to the practices through which people develop, deploy, and use AI (Ruster & Daniell, 2025).

Research on AI governance operates across multiple scales and levels of analysis, including policy, technology, and organizational practice (Attard-Frost & Lyons, 2025; Batool et al., 2025). Policy-level work examines how governments and international bodies regulate AI development and deployment, including efforts to protect privacy and prevent discriminatory outcomes (e.g., Dafoe, 2023; Schiff et al., 2021; Ulnicane et al., 2021). Technology-level approaches address how systems are designed and evaluated, with particular attention to fairness, explainability, and the data and measurement choices that shape model performance (e.g., Jacobs & Wallach, 2021; Mehrabi et al., 2021; Mittelstadt et al., 2016). Organizational-level work considers how firms integrate AI into their operations and distribute responsibility for its use, providing conceptual foundations that continue to be developed (Bankins et al., 2024; Birkstedt et al., 2023; Cihon et al., 2021).

Current research presents organizational AI governance as an arrangement of decision rights and practices spanning the AI lifecycle (Krijger et al., 2023; Mäntymäki et al., 2022; Schneider et al., 2023). Structural, relational, and procedural mechanisms take concrete form in arrangements such as algorithm review boards (Hadley et al., 2025; Papagiannidis et al., 2025). Translating these prescriptions into everyday work, however, remains challenging. Evidence from real-world implementations shows that the responsible AI practices organizations enact often diverge from those recommended in principle (Bach et al., 2025). Conventional IT governance provides a

starting point, but AI also requires attention to changes in model behavior and organizational authority (Berente et al., 2021). For example, emerging AI models may lose reliability as their data change, while the workflows built around them can make responsibility harder to locate (Kellogg et al., 2020; Schuett, 2024; Sculley et al., 2015).

Existing literature already differentiates AI governance by risk, context, and lifecycle stage. The National Institute of Standards and Technology (NIST) AI Risk Management Framework, for example, connects system inventories and ongoing monitoring to assigned responsibilities and organizational resources (NIST, 2023). Organizational accounts also recognize that AI governance depends on context, meaning the same system may require different oversight mechanisms depending on where it is deployed, who relies on it, and what is at stake. Recent work therefore recommends differentiated oversight of individual AI modules according to the characteristics of the system in use (Mikalef et al., 2025). Building on this work, we examine how learning after deployment and the delegation of consequential action create different governance demands, and how organizations can respond across a portfolio of interdependent systems.

AI systems have qualitatively different capabilities and organizational consequences (Chalmers et al., 2026; Kemp, 2024). Adaptability and autonomy are two of the characteristics along which AI systems are known to differ, and both are already recognized in definitional frameworks and scholarship (OECD, 2024; European Parliament and Council of the European Union, 2024). Recent frameworks track decision authority, process autonomy, and accountability across changing governance thresholds (Engin & Hand, 2026), and treat autonomy as graded by connecting combinations of agent properties to governance responses (Kasirzadeh & Gabriel, 2026). These antecedents establish the value of dimensional analysis.

Our contribution in this article is more specific and directed toward what these dimensions mean for organizational governance of AI, particularly how adaptability and autonomy alter governance within an organizational portfolio, including the conditions under which their interaction becomes recursive and shared dependencies affect the allocation of oversight. Our organizational focus concerns how work practices produce the evidence used to assess AI and distribute authority to act on it. The same technical system can require different governance arrangements when its recommendations, learning data, or permissions are embedded in different routines.

We address this problem by developing a two-dimensional model organized around adaptability and autonomy. We draw on the concept of *adaptability* to describe changes in system behavior produced by learning after deployment. This focus is consistent with research that differentiates

AI systems by their ability to learn and adapt to new information or environments (e.g., Jarrahi & Glaser, 2025; OECD, 2024). For example, a model trained once can encounter a changing environment without learning (Kühl et al., 2022). We use the term *autonomy* to describe the extent to which a system selects or executes consequential actions without contemporaneous human direction (Kasirzadeh & Gabriel, 2026; Parasuraman et al., 2000). The selection of a course of action and its implementation can be delegated to different degrees (Parasuraman et al., 2000). Even a nominally advisory system can acquire practical authority when its recommendations go unchallenged. This distinction matters because adaptability changes how long a specific governance approach remains applicable, whereas autonomy changes where consequential action occurs and how quickly an organization can intervene.

By proposing the model, we make four related contributions. First, we explain how two established system characteristics challenge different assumptions underlying organizational oversight. Second, we clarify their boundaries by distinguishing adaptability from environmental drift and examining autonomy through both formal permissions and practical authority. Third, taking an interactional approach, we direct attention to how governance demands can compound when system action shapes the evidence used for subsequent adaptation, making monitoring and action controls interdependent. Finally, we extend this analysis to portfolio decisions about ownership, shared controls, and continuous reassessment. In the rest of this paper, we develop the model, illustrate it with a hospital AI portfolio and examples from other sectors, and discuss its ethical and organizational implications.

2 A TWO-DIMENSIONAL MODEL FOR DIFFERENTIATED AI GOVERNANCE

We define organizational AI governance as the sociotechnical arrangement of decision rights and oversight practices through which organizations pursue the benefits of AI while addressing its risks and ethical harms. Drawing on extant literature, we counterpose adaptability and autonomy, resulting in an interactive typology that directs attention to where governance challenges intensify and what form they may take. In our formulation, neither dimension yields a fixed governance formula; sector-specific obligations and the scale and reversibility of possible harm remain pivotal. The dimensions help identify when organizations should reconsider reliance on stable system behavior or contemporaneous human control. This focus draws on the distinction between

learning and execution (Kühl et al., 2022) and on research showing how agency attributed to AI can alter practical control and responsibility (Leonardi, 2025).

We developed the model through a focused integrative synthesis of research on organizational AI governance and the sociotechnical dynamics of AI use, including learning and drift, human control, and algorithmic accountability (e.g., Faraj et al., 2018; Stelmaszak et al., 2026; Wieringa, 2020). We selected adaptability and autonomy because, in our analysis, each challenges a different assumption on which oversight may rely: (a) the governed system remains stable, or (b) humans retain contemporaneous control over consequential actions. These assumptions direct attention to the form of oversight. Impact, reversibility, and severity still shape how much oversight a system needs, but they do not determine its position in the matrix comprising these two dimensions, which depends on post-deployment learning and delegated action.

Table 1 locates lower, intermediate, and higher positions on each dimension. Because our primary objective is to develop these distinctions, we present a conceptual synthesis rather than a systematic review. We use documented cases and illustrative examples to clarify where the constructs begin and end and what governance implications may follow.

Table 1. Two dimensions of AI system characteristics

Dimension	Lower end of the dimension	Intermediate conditions	Higher end of the dimension
AI Adaptability <i>(how much the system changes through learning after deployment)</i>	The model's learned parameters or decision rules remain fixed. <i>Governance stake:</i> earlier assurance may remain applicable, but changes in users, data, or context still require review.	The system is retrained on a set schedule, with each new version checked before it goes live. <i>Governance stake:</i> assurance must be renewed with each release; the organization governs change, but only at defined moments.	The system keeps changing on its own between reviews as it learns from new data or users' interactions. <i>Governance stake:</i> system behavior can move beyond the last approval, so monitoring must follow change and trigger timely review.
AI Autonomy <i>(the extent to which the system acts without a</i>	The system advises; a human makes the consequential decision and carries it out.	The system proposes an action a person confirms, or carries out a narrow, reversible action on its own.	The system selects and takes consequential, often multi-step action with little real-time human involvement.

<i>person in the moment)</i>	<i>Governance stake:</i> accountability rests with an identifiable human decision-maker.	<i>Governance stake:</i> authority is shared, so the organization must define what may proceed without sign-off and ensure human review is thorough, rather than nominal.	<i>Governance stake:</i> the system's action can precede review, so authority, intervention rights, and mechanisms for recourse must be set in advance.
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2.1 Conceptualizing AI Systems along Two Dimensions of Shift

2.1.1 AI adaptability: *The shift from stable to evolving systems*

AI adaptability refers to the degree to which a deployed system modifies its behavior based on learning from new data, feedback, or environmental interaction (OECD, 2024; Queiroz et al., 2024). This distinguishes learning systems from fixed models whose performance dwindles over time because the environment shifts away from its training conditions (Gama et al., 2014).

At one end of the continuum are relatively stable systems, including rule-based systems and fixed-weight models whose logic remains constant until developers release replacements. Intermediate arrangements allow retraining at set intervals, with validation before each release. At the other end, systems update through automated batches or online learning. Based on our definition, reinforcement learning counts as post-deployment adaptability only if learning continues during operations; the training technique alone does not establish adaptability. What changes the governance approach is how much the system can change before the organization evaluates it, and whether monitoring and approval processes keep pace with that change (Minkinen et al., 2022).

This shift in adaptability affects the temporal demands of governance. A system's behavior after an update may differ from the behavior that justified its approval and use in specific organizational scenarios. Performance can decline because the model has changed or because the environment around it has shifted. Organizations therefore need to diagnose the source of change before acting. For example, if a recent update to the model has led to worse performance, the governance response may be to revert to the previous version or repair the new one. If the model was left unchanged but tasks have shifted over time, governance should focus on whether the model and its training data still fit the current organizational conditions.

As a system changes, the organization's understanding of it can fall behind its current behavior. When retraining or continued learning outpaces documentation describing the system, people

working with it may struggle to explain its behavior or justify its continued use (Ananny & Crawford, 2018). We therefore treat adaptability and opacity as related but distinct concerns. A system that changes through a managed, well-documented process can remain more accountable than a static one the organization never effectively understood in the first place. Governance depends on whether the organization can explain how the system currently works and justify its continued use. A key issue is temporal lag, which refers to an outdated assessment that may no longer support the decisions made by the version in use today.

2.1.2 AI autonomy: The shift from decision support to delegated action

AI autonomy refers to the degree to which a system selects or executes consequential organizational action without contemporaneous human direction. Autonomy varies by degree (Kasirzadeh & Gabriel, 2026; OECD, 2024). Parasuraman et al. (2000) distinguished decision and action selection from action implementation. Automatic execution alone therefore does not indicate how much discretion has been delegated. For example, a system may implement a tightly prescribed action, while an agent can decompose a larger task and use external tools to pursue a broader goal (Kolt, 2026). The distinction concerns the discretion exercised in use; a rule-based design alone does not establish low autonomy.

At the lower end, decision-support systems provide recommendations or analyses while humans retain responsibility for consequential choices and their implementation. Intermediate cases include systems that select an action for human confirmation or automatically implement a narrow, reversible rule. Higher autonomy involves selecting and executing actions across wider-reaching or multi-step tasks with limited real-time involvement. But how much authority a system is formally granted is only part of the equation. Even when a human operator makes the official decision, in practice that human may defer to the system and approve its outputs as a matter of routine rather than subject them to genuine review (Engin & Hand, 2026). Heavy workloads or limited expertise can contribute to formal human decision rights becoming largely ceremonial (Green, 2022).

As autonomy increases, accountability becomes harder to anchor in frontline human discretion. When action precedes meaningful review, organizations must establish who authorized it, who could have intervened, and who owes an account to those affected (Bovens, 2007; Novelli et al., 2024). Merely ensuring human involvement somewhere in the process does not resolve these concerns (Robbins, 2024). Organizational actors need enough time and expertise to thoroughly

examine the system, and the authority to overturn outcomes, if necessary. Affected stakeholders also need a way to contest decisions autonomously or semi-autonomously made by AI.

Figure 1 summarizes these differences using a coarse 2×2 representation of the continuous dimensions in Table 1. The quadrants indicate broad differences in the form of oversight; intermediate systems would not be plotted at the extremes of either dimension. For relatively stable decision-support systems in the lower-left quadrant, established IT governance practices, including validation and periodic review, are most likely to remain viable. Increasing autonomy makes the authorization and interruption of action more consequential; increasing adaptability requires monitoring and validation to adapt to system change. When both are high, governance demands can compound if system action becomes connected to adaptation. A system may alter the observations used for its next update, leaving an organization to evaluate changing behavior against evidence that the system itself helped produce. Figure 2 in Section 2.5 illustrates this possible feedback pathway and the questions that must be answered to identify it.

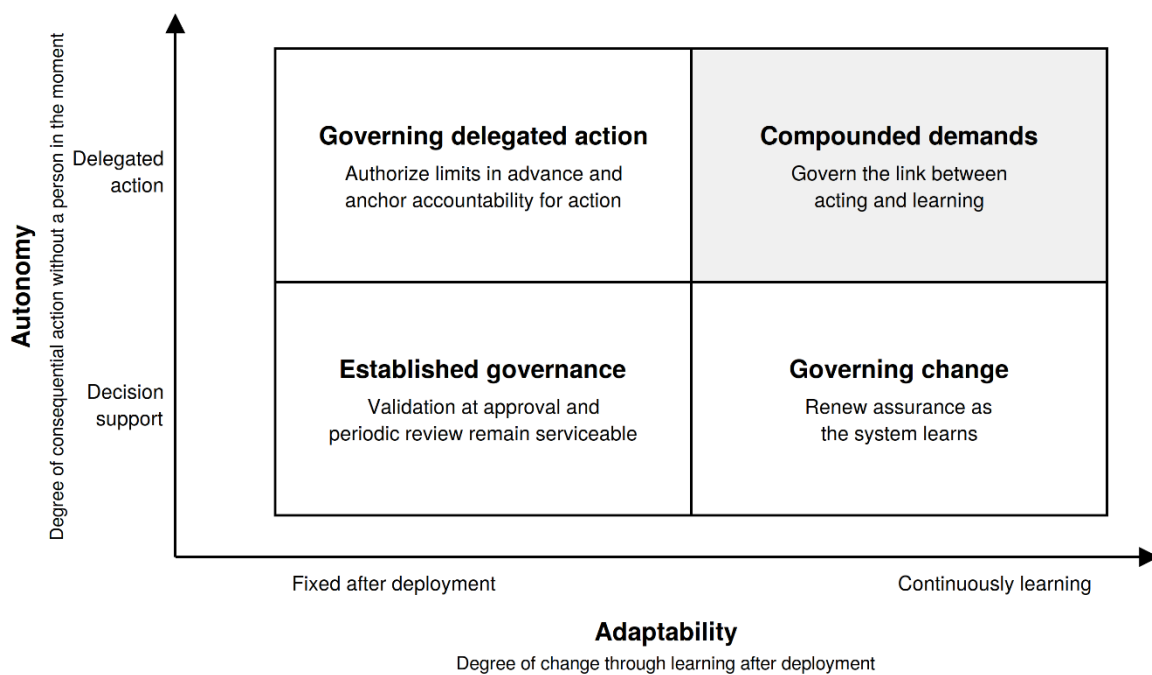


Figure 1. Differentiated governance demands across adaptability and autonomy.

2.2 Differentiating AI Governance based on Adaptability and Autonomy

Organizations use various systems simultaneously, so an organization's AI portfolio may include a stable decision-support tool alongside a retrained recommender or an agentic system that carries out several steps toward a goal. Their requirements for validation and intervention differ, even within the same business function (Chalmers et al., 2026; Riemer et al., 2026). A differentiated approach may accommodate each system according to how it changes through learning and the extent to which it selects or executes consequential action without human direction at the time of action.

Our two-dimensional model makes these questions explicit. Organizations can use the model to decide when to renew assurances as a system updates, where to build intervention into a workflow as actions become more autonomous, and when to link monitoring with action controls because the system's actions shape subsequent learning. We illustrate this using the example of a hospital network deploying four AI systems, drawing on additional examples to show how the same distinctions travel across sectors.

A system's position on the adaptability and autonomy dimensions helps organizations decide what oversight mechanisms it needs. Learning after deployment, for example, calls for reviewing changes in system behavior, while greater autonomy calls for clear limits on action and designated people who can intervene. These arrangements must also meet the legal requirements of the sector and provide protection proportionate to the extent of possible harm to different stakeholders (Sargeant, 2023; Schuett, 2024). For example, even a fixed system with limited autonomy could require substantial safeguards if its decisions could seriously affect health or access to credit (NIST, 2023).

Table 2. Illustrative example: Classifying a hospital's AI portfolio based on adaptability and autonomy

Dimension	Lower Autonomy	Higher Autonomy
Lower Adaptability	Fixed-Weight Surgical Scheduling Recommender: ranks operating room allocations for human coordinators	Model-Based Prior Authorization Engine: uses a stable trained model to approve, deny, request additional evidence, or

	using historical utilization, case mix, and staffing data; parameters remain fixed after deployment. Salient governance challenge: baseline oversight of a stable decision-support system.	flag for review under a delegated mandate. Its discretion concerns the route taken within policy-defined limits, although its learned parameters remain fixed. Salient governance challenges: delegated action, accountability, and recourse mechanisms.
Higher Adaptability	Diagnostic Imaging Aid: flags possible abnormalities in scans; retrained quarterly and is deployed only after version-specific validation; clinicians retain final judgment. Salient governance challenges: performance and subgroup changes across versions.	Continuously Updating Sepsis Prediction Agent: monitors ICU data, updates automatically as data and feedback accumulate, and triggers protocols or escalation pathways. Salient governance challenges: action-data feedback, delegated action, and time-sensitive intervention.

Note: Placement reflects formal delegation and relative differences in updating: the aid that is retrained quarterly is more adaptive than the fixed models but less adaptive than the continuously updating agent. Practical authority can also shift through human deference to the system, as discussed in Section 2.4. The examples are illustrative configurations, not particular deployed products.

Oversight adequate for the scheduling recommender would leave important aspects of the sepsis agent unchecked. The diagnostic aid places particular demands on validation after retraining, while the prior authorization engine raises questions about who authorizes action and how affected people can contest it. Similar differences arise in finance. A fixed underwriting score may require closer monitoring as borrower populations change, although the model is not adaptive. For a fraud system that places holds automatically, the organization must also limit which payments can be stopped and provide timely correction before a legitimate customer bears a lasting cost. The compounded case connects change and action in ways that controls designed specifically for either application may miss.

2.3 Governing AI When Adaptability Increases

Retrained and continuously updating systems require different monitoring and approval routines than systems that remain fixed, even when their performance indicators look similar (Minkkinen et al., 2022). For a fixed model, the first question is whether the cases it analyzes or the work

practices surrounding its use have changed since it was last evaluated. For a periodically retrained model, governance must also examine each new version prior to implementation. Online learning introduces a further challenge because the system's behavior can change between formal reviews.

Drift must therefore be understood in relation to organizational use. Statistical or operational drift occurs when the data a system encounters or the way it is used changes, affecting its performance and the kinds of errors it makes (Gama et al., 2014). We use normative drift to refer to cases in which the system's behavior or effects deviate from the organization's stated values or obligations, even when aggregate performance measures remain satisfactory (Jacobs & Wallach, 2021; Wieringa, 2020).

To illustrate this phenomenon, we build on Grisold et al.'s (2025) example of a warehouse in which an AI-based agent gradually reduces its desired distance to human workers as operations become more efficient, potentially undermining a safety-oriented norm. Extending this example, an organization could compare separation distances and near misses against throughput across system versions, and examine workers' reports of evasive action. Rising throughput accompanied by more frequent safety interventions would prompt investigation of whether learning had favored speed at the expense of worker safety. The manager responsible could restrict the agent's operating area or pause updating while the team assesses its learned behavior and evaluation criteria. Please note that these are proposed detection and response steps, rather than evidence from a field-validated drift episode.

Predictive policing presents a similar pathway. Mathematical modeling and simulations informed by data from Oakland, CA show how patrol allocation can concentrate observation in selected areas, with newly recorded incidents reinforcing subsequent allocation decisions (Ensign et al., 2018). The analysis identifies a governance problem: system performance may look satisfactory when assessed against data shaped by its own decisions. More recorded incidents may reflect more intensive observation, even without a corresponding increase in underlying harm. To address this problem, organizations must compare recorded incidents with other evidence of harm that is less influenced by where patrols are sent. They should also examine whether repeated surveillance is concentrated in particular neighborhoods or among particular groups. If the increase in recorded incidents mainly results from more intensive observation rather than

independent evidence of greater harm, retraining should pause until this feedback problem is addressed.

In the hospital's AI portfolio, the diagnostic imaging aid might be retrained quarterly, but clinicians make final judgments. Each version therefore needs to be compared with its predecessor, including differences in subgroup performance and use across sites. Similarly, during a review process at Revenue NSW in Australia, the existing model remains in service while newly retrained versions are evaluated. The existing model is replaced only after a new version achieves higher accuracy with an acceptable level of explainability (Smith et al., 2026). Learning does not continuously change the model already in service and each new version is subjected to an approval process prior to implementation. Although a human decision-maker is involved in the diagnostic process, there is still a need to validate the model that produces the outputs subjected to human judgment.

One of the key issues when systems become more adaptable is that explanations and documentation associated with an earlier version may cease to describe current behavior after retraining or a change in deployment conditions (Ananny & Crawford, 2018). Typically, organizations should be able to identify the evidence supporting an approved version and restore a previous version if inconsistencies or irregularities emerge. In a Canary Islands public administration system for classifying and summarizing legislative documents, maintaining identifiable model versions supports comparison, auditing, and rollback (i.e., a return to an earlier version), while legal experts curate data and review outputs (Aguilar et al., 2026). An organization should record and review changes to a system over time, even when human officials retain authority over policy decisions. More frequent updating requires correspondingly frequent checks on whether the evidence still justifies the use of the system (Minkinen et al., 2022; Mökander & Floridi, 2023).

Because adaptive AI systems can change after deployment, governance must continuously adapt. Organizations need to monitor technical performance, interpret what observed changes mean, and adjust their oversight practices when necessary (Ibitoye et al., 2025; Minkinen et al., 2022; Raji et al., 2020). For example, after retraining a system, an organization concerned about normative drift could examine whether some groups have become more likely to receive adverse decisions. It could also assess whether the criteria guiding those decisions still reflect its commitments, using evidence that was not produced by the system's own decisions where possible. As a result,

material divergence from goals requires an organizational response, such as investigation or a pause in retraining, with a designated person authorized to restore an earlier version or suspend the system (Aguilar et al., 2026; Rakova et al., 2021).

2.4 Governing AI When Autonomy Increases

The shift from decision support to delegated action changes how accountability must be organized. AI governance must address both how people use recommendations and the scope of action granted to the system as autonomy increases (Leonardi, 2025; Novelli et al., 2024; Tsamados et al., 2025). Principal-agent concepts help explain this difficulty: a person or organization delegates a task to an agent, but may be unable to determine whether the agent remained within its mandate, especially when the goal is broad and the means of pursuing it are left unspecified (Jarrahi & Ritala, 2025). A narrow rules engine executes predetermined actions, whereas a more autonomous AI agent may decompose a broader task and use external tools to pursue it (Kolt, 2026). As more actions occur before human review, organizations need to define the limits of this discretion and assign responsibility for determining whether the system remains within them (Cobbe et al., 2023; Novelli et al., 2024; Schneider & Symeonidis, 2026; Wieringa, 2020).

At lower levels of autonomy, clear decision rights and review routines can keep accountability anchored in human discretion (Kirsch, 1996; Parasuraman et al., 2000). Practical authority may nevertheless shift without formal delegation when human reviewers routinely accept a system's recommendations. Treating AI as an independent actor may further reduce reviewers' sense of control and responsibility, even though formal accountability remains with them and the organization (Leonardi, 2025). Organizations should therefore examine whether reviewers genuinely question the system's output, whether they can choose a different course of action, and whether records show what the system recommended and how the reviewer responded (Green, 2022).

The hospital portfolio example makes this distinction more concrete. An AI scheduling recommender formally leaves allocation decisions to a human coordinator. If coordinators

routinely accept its recommendations, however, the system gains practical authority even though decision-making authority has not been formally delegated to it. A prior authorization engine presents a different arrangement because it is formally authorized to approve or deny requests within specified limits. A fraud detection system that automatically places holds on accounts provides another example of formally delegated action. However, the organization needs to specify which holds may proceed without human approval and who can promptly correct an erroneous one (Alfrink et al., 2023; Kroll et al., 2017). Such systems also strain established controls when organizations cannot supervise or ascertain every step involved in producing an organizational outcome (Kirsch, 1996; Ouchi, 1979). Instead, organizations may rely heavily on measurable results, such as the number or value of improper payments prevented, to judge performance. This involves an inevitable trade-off. Making it easier for the system to flag a payment may improve this measure by blocking more suspected fraud, but it may also increase erroneous holds on legitimate payments. The holds take effect immediately, whereas their costs may only become visible later through human reviews and appeals. AI governance therefore should consider both the amount of fraud prevented and the consequences of erroneous holds for affected people.

Responsibility for governing more autonomous systems often extends across the people who develop, deploy, and oversee them (Grote et al., 2026), creating a need to clarify and assign accountability and intervention authority prior to system deployment (Shavit et al., 2023; Zhu et al., 2026). Relevant stakeholders who work alongside the AI system should know who can pause operations, who is responsible for investigating alerts, and who is able to explain outcomes and provide recourse to affected people. Transparency also requires people and routines that make explanations usable. HireVue, an AI-based interview-assessment system, provides a relevant example. Records of model design and training were combined with explanations developed by data scientists, occupational psychologists, and client-facing staff. Tarafdar et al. (2025) reported that when recruiters questioned particular scores, occupational psychologists reviewed sample interviews with clients to compare the behaviors displayed by candidates with the criteria used to produce the scores. This process helped clients understand how the system arrived at its assessment and provided a basis for examining disputed results.

As autonomy of an AI system increases, organizations must ensure that intervention authority of human counterparts can be effectively exercised in practice. Responsible individuals need to have the necessary expertise, information, time, and organizational standing to alter an outcome (Cavalcante Siebert et al., 2023). Rehearsing failures can reveal whether those responsible can make a decision, halt or isolate the system, and return it to a known safe state before the consequences become difficult to reverse. Restoring a previous system version cannot undo a denied service or another harm already experienced, so responsibility for correction and recourse must remain clear.

The stakeholders tasked with these responsibilities also require organizational support. A review of 224 responsible AI tools shows that support is directed primarily to designers and developers, whereas organizational leaders, deployers, end-users, and affected communities receive less support. The review also identifies limited documentation showing whether these tools have been validated (Kuehnert et al., 2025). Assigning responsibility should therefore include determining whether relevant stakeholders have access to practical resources and tools rather than assuming that one governance toolkit will address all needs. Increasing autonomy changes where action occurs and where accountability must be anchored: those answerable for outcomes need practical authority to constrain and interrupt the actions producing them. Table 3 brings these requirements together with the capacities necessary to address adaptability, compounded effects, and portfolio interdependence.

2.5 Intersections and Compounded Governance Challenges

Our distinction between adaptability and autonomy is analytical; in practice, these dimensions can co-occur. Their governance demands compound when monitoring adaptation depends on controlling action, or vice versa, so the two sources of complexity cannot be evaluated and addressed independently. This interaction extends our model beyond a typology. Its significance depends on how learning and action become connected in use.

A key sociotechnical dynamic of compounding is a recursive feedback mechanism between learning and action (see Figure 2). Autonomy contributes to the action side of this relationship because the system selects or executes an intervention that can change its environment.

Adaptability contributes to the learning side because observations made after the intervention are used to update the system. When these processes connect, the act of exercising autonomy changes the evidence from which the system learns, while adaptation changes how it subsequently exercises autonomy. Governance of action and learning can therefore no longer proceed independently. High adaptability and high autonomy create the potential for compounding, but recursive compounding occurs only when three links are present: (a) the system's actions shape what is subsequently observed, (b) those observations are used to update the system, and (c) the updated system changes subsequent actions.

Performative prediction provides a formal account of how deployment can change the distribution on which later predictions depend. A prediction can influence a decision, and acting on that decision can change the outcomes and data the system later observes (Perdomo et al., 2020). Predictive policing simulations make the full cycle more visible. When a model directs more patrols to particular areas, more incidents may be discovered and recorded there. Those records are then used to update the model, which influences subsequent patrol allocation decisions (Ensign et al., 2018). This analysis illustrates the feedback mechanism without establishing the degree of autonomy in a particular deployment. Consistent with Section 2.1.2, human involvement does not necessarily mitigate practical authority when recommendations routinely determine action; that relationship must be established in the setting under review. Article 15(4) of the EU AI Act recognizes a related potential pitfall for high-risk systems that continue learning after deployment, specifically when biased outputs influence inputs used in future operations (European Parliament and Council of the European Union, 2024). These accounts explain the recursive mechanism. In our model, compounding between high adaptability and high autonomy occurs when the system also exercises sufficient formal or practical authority over actions that shape its future learning evidence.

These links are organized through the routines in which AI operates. An assemblage perspective locates agency in relations among data, algorithms, decisions, and routines (Glaser et al., 2024). For AI agents, espoused values are transformed through configuration, embedding, and performance, with enacted outcomes informing later configuration (Glaser et al., 2026). Applied to our model, work routines determine which recommendations become actions; recording and labeling practices determine which consequences become learning evidence; and release arrangements determine whether an update changes subsequent action. Governance must therefore examine how these practices connect, including whether success criteria make harmful consequences difficult to see. This broader organizational feedback constitutes recursive compounding in a narrower sense only when it includes post-deployment learning that changes the system's subsequent actions.

The continuously learning sepsis prediction agent in the hospital portfolio provides another illustrative example. The algorithm updates based on patient data while triggering protocols that influence care. If those interventions shape the outcomes used in later updating, the system partly produces its own learning environment. An adaptive fraud system could create a similar recursive loop when automated holds influence which transactions are completed and therefore available for later annotation and retraining. When the system blocks a transaction, the organization may never learn whether it was actually fraudulent or legitimate. Unless a human-centric review resolves this uncertainty, the transaction cannot provide a reliable label for retraining.

Mapped onto Figure 2, the system's decision to trigger a care protocol or place a transaction hold is the action. The patient outcome or the later determination that a transaction was legitimate or fraudulent is the observation. Using those observations to retrain the system constitutes the update. This may become a governance issue because the action may change what can later be observed. The updated system may therefore learn partly from conditions created by its previous action rather than from independent evidence of the underlying clinical or fraud risk. The sepsis and fraud configurations are illustrations of the proposed mechanism. Formal models and simulation studies show how these links can produce recursive feedback in specific conditions and help identify what governance would need to focus on, but they do not clarify how often complete loops occur in deployed organizations or whether the proposed controls work in practice. Answering those questions requires empirical study of actions, updating, and organizational practice over time.

Once this recursive pathway is identified, organizations can monitor and address the adverse consequences of compounding by reducing one of the two dimensions or separating the processes that connect them. Temporarily suspending updates could remove adaptability from the loop and break the connection between observation and updating. The system may continue to act, but new observations do not change the deployed model. Independent comparison evidence can reveal how the system's earlier actions influenced what was observed. The connection between action and observation is broken only when the evidence used to update the system is not affected by those actions. Using the system as an advisor can diminish its autonomy in the loop because the AI recommends action rather than carrying it out. This distinction matters only when humans "in the loop" are given opportunities to independently evaluate system recommendations and choose other courses of action. If they routinely and uncritically accept recommendations, the system may still shape subsequent actions as well as the evidence available for later learning. A release gate constrains the connection between updating and subsequent action by requiring a new version to be reviewed before it is implemented. It creates an opportunity for review, but does not remove feedback already present in the retraining data.

Figure 2 summarizes how autonomy enters the loop through action, and adaptability enters through post-deployment updating. The process depicted in this figure can serve as an AI governance diagnostic approach when it comes to compounding effects. Organizations should ask whether the system's actions shape later observations, whether those observations inform an update, and whether the update changes subsequent action. Three affirmative answers identify a recursive pathway requiring coordinated oversight of action and learning. When an answer is unknown, the organization needs further evidence before treating action and learning as independent.

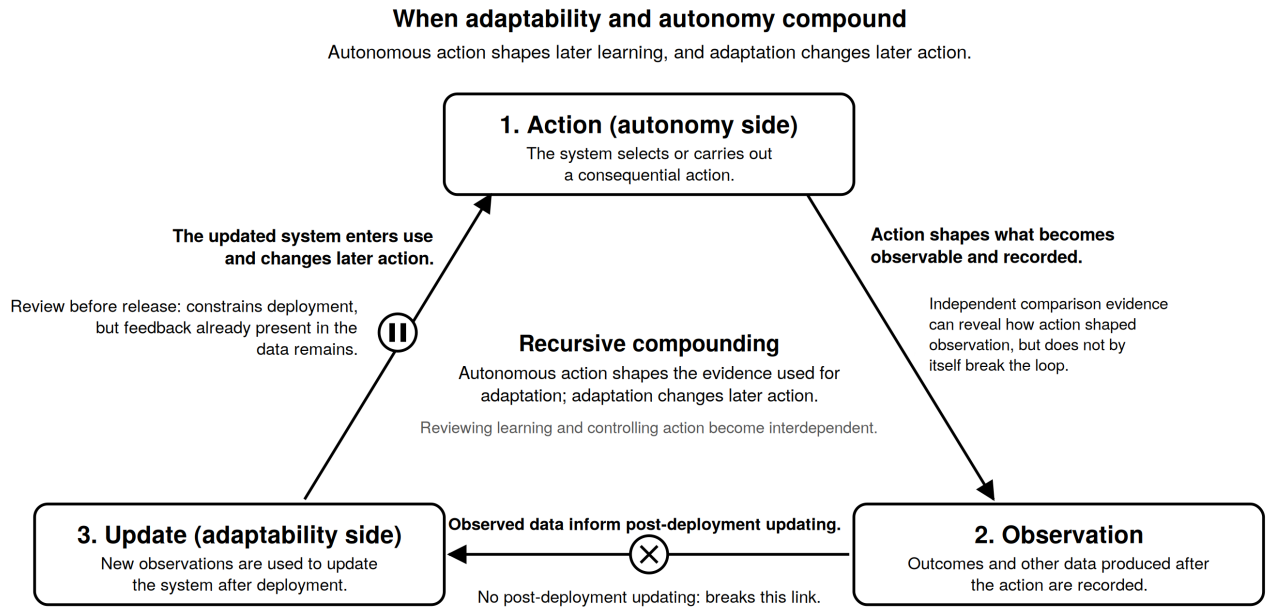


Figure 2. Recursive compounding through action, observation, and post-deployment updating.

Once a recursive loop is identified, the organization must determine whether an apparent improvement reflects better decisions or a change in what the system’s actions made visible. This requires documentation linking each system version to its actions and the outcomes that informed subsequent learning. In the example of policing, recorded incidents should be compared with other evidence of harm that was less influenced by where police patrols were sent. Such a comparison can help distinguish a change in underlying harm from a change produced by concentrating observation in particular areas.

Because the loop spans both action and learning, responsibility for assessing it must be matched with authority over both processes. Pausing action may be insufficient if observations produced by earlier actions continue to feed retraining, whereas suspending updates may allow harmful actions to continue. The organization must therefore identify who can interrupt each process, who assesses the connection between them, and who decides when action or updating may resume.

We use the term *oversight capacity gap* to refer to situations in which committees and review procedures remain formally in place, but reviewers lack the resources or authority to act. A formal requirement for human oversight may fail in practice (Green, 2022); meaningful human control requires substantive conditions beyond that formal requirement (Cavalcante Siebert et al., 2023). This gap may emerge when the volume of updates or speed of action exceeds the capacity allocated to review, particularly when consequences occur before anyone can intervene. We propose that longer review queues, unresolved exceptions, and patterns of routine approval or limited override can serve as warning signs of weakening oversight. Table 3 summarizes the governance implications of increased adaptability, increased autonomy, their recursive interaction, and portfolio interdependence.

Table 3. Governance implications of increased adaptability, increased autonomy, recursive compounding, and portfolio interdependence

Governance condition	What changes	Governance assumption challenged	Governance implications
Increased adaptability	System behavior changes through periodic retraining or ongoing learning after deployment	Earlier validation and documentation may no longer describe the deployed version	Assign responsibility for maintaining current assurance evidence; identify and compare versions; match monitoring and approval routines to the frequency of system changes; retain the ability to restore an earlier version
Increased autonomy	The system selects or executes consequential actions with less contemporaneous human direction	Human review may occur after consequences emerge, and practical authority can migrate to the system even when humans retain formal accountability	Authorize and enforce action limits in advance; ensure human reviewers can question or override the system; specify who can pause action; retain records for investigation and explanation; establish protocols for contestation and correction

Recursive compounding	The system takes actions that shape observations used for updating, and the updated system changes subsequent actions	Learning and action can no longer be monitored and controlled independently	Assess whether apparent improvement reflects action-shaped evidence; connect records of system versions, actions, and outcomes; compare with evidence less dependent on the system's actions; assign authority to interrupt both action and updating; ensure responsible roles have sufficient time, information, and authority
Portfolio	Applications share models, data, vendors, or workflows, so a change can affect several systems	Separate application reviews may miss shared dependencies and their consequences	Trace dependencies; assign application owners and responsibility for shared components; reuse common controls where appropriate; coordinate reassessment when a shared component changes or fails

3 DISCUSSION AND IMPLICATIONS FOR AI GOVERNANCE RESEARCH

Existing frameworks recognize that organizational AI governance needs to account for differences among systems and their uses. We contribute by explaining how adaptability and autonomy change the evidence and authority required for oversight, and when their interaction creates interdependence between monitoring and intervention. This focus connects responsible AI principles to operational choices while leaving room for contextual risks that the two dimensions do not capture (Batool et al., 2025; Morley et al., 2020).

Our interaction analysis identifies further complexity. Recursive compounding occurs when actions shape observations used in subsequent learning and updating leads to changed actions. Responsible integration requires the latitude to question that evidence, compare it with information that is less dependent on the system's decisions, and reconsider what counts as acceptable performance (Ananny & Crawford, 2018; Jacobs & Wallach, 2021; Lebovitz et al., 2022). Such judgment becomes especially important when efforts to improve performance measures conceal changes in who bears the costs of the system's decisions.

The model also treats AI governance as an organizational capacity sustained across the lifecycle (Akbarighatar, 2025; Berente et al., 2021). This illustrates why responsible use requires exercising judgment at both individual and organizational levels rather than mechanically applying assessment tools (Sturm et al., 2026). Our typology based on adaptability and autonomy reveals the need for people who can question a system's outputs under these conditions and act on exceptions. Algorithm review boards and healthcare governance programs provide institutional settings for such work (Hadley et al., 2025; Kim et al., 2025). Their review cadence and intervention authority need to match the complexity of the systems they oversee.

We argue that the dimensions of adaptability and autonomy underscore the importance of human agency in AI-mediated organizing. Adaptive systems require people to interpret changes and decide whether continued use remains justified. Governing autonomous systems requires people who can define and interrupt action pathways. These responsibilities connect human-AI collaboration with algorithmic management because the same person may use AI to manage others while also being managed through it (Hillebrand et al., 2025). The relationship between automation and augmentation remains an organizational concern (Raisch & Krakowski, 2021). Conceptually, patterns of human improvisation can reveal gaps between espoused and enacted values (Glaser et al., 2026). For governance, recurring overrides and workarounds therefore warrant examination as possible evidence that evaluation criteria or delegated permissions need to be revised. People must have access to the relevant records and the authority to subject such patterns to review.

We also highlight that employees operating several AI agents may need to engage in strategic direction, coordination, and task-level supervision, taking on responsibilities resembling different levels of management (Van Quaquebeke et al., 2026). We view this as a possible future development rather than evidence that such arrangements are already widespread or effective. This possibility has implications for job design and employee training: organizations may need to clarify these new supervisory responsibilities and prepare staff to set directions for agents, coordinate their activities, and intervene when necessary.

3.1 Coordinating Governance Across AI Portfolios

Our framework positions individual AI systems along the dimensions of adaptability and autonomy, but organizations rarely govern these systems in isolation. Several AI applications may depend on the same model, data source, vendor, or organizational workflow. A change or failure in a shared

component can therefore affect systems occupying different positions in the framework, including systems that were reviewed separately (Hudig et al., 2026).

Portfolio interdependence concerns dependencies across multiple AI systems. It can intersect with recursive compounding when an adaptive model or service is shared across applications: feedback generated through one use may influence an update that subsequently affects the others. This possibility requires organizations to connect application-level oversight with responsibility and coordination across the portfolio.

System inventories and resourcing aligned with risk priorities provide an organization-wide baseline (NIST, 2023). Building on that baseline, the portfolio approach assigns each system an accountable owner and records dependencies shared with other applications (Cobbe et al., 2023). Architecture mapping and responsibility walkthroughs can document how components connect and establish who is responsible when they change or fail (Mikalef et al., 2025). For example, when several applications depend on the same model or data pipeline, a responsibility walkthrough would identify which applications could be affected by a model update or data problem and who owns each one. It would also clarify who evaluates the consequences within each application, who decides whether its use should continue, and who coordinates the response across the portfolio.

Such a portfolio approach becomes concrete in two related governance tasks: tracing incidents across shared services and coordinating responsibility across organizational levels. First, records that track agents and their activities across services can help owners identify how incidents involving shared components have affected several applications (Chan et al., 2024). This information supports coordinated decisions about which systems require investigation or temporary restrictions. Organizations may reuse common controls for a shared platform, but local owners must still examine its performance, consequences for affected groups, and potential recourse options within each application. At the portfolio level, review capacity should be concentrated where updates are implemented without prior approval, actions are difficult to reverse, or shared dependency exposes several systems at once.

Affordability places a practical boundary on these arrangements, especially for smaller organizations. Shared inventories, platform assessments, and incident procedures may reduce duplicated work across applications, even though each application still needs to be assessed to identify local effects and options for recourse. This is a potential economy of shared governance, not a demonstrated cost savings. When adequate review and intervention cannot be resourced, it

may be more defensible to limit autonomous action, require approval before updates are implemented, or defer deployment than to maintain nominal oversight.

Second, portfolio governance requires a workable division of responsibility between central governance resources and local application owners (Viaene, 2026). A qualitative assessment across selected business units found fragmented oversight and unclear escalation paths. The resulting ARGO framework combines shared standards and central advisory resources with local responsibility for monitoring, review, and incident reporting (Jafari Meimandi et al., 2025). Applied to our portfolio approach, a central team could identify all applications affected by an incident involving a shared model and coordinate the overall response, while local owners assess the consequences within particular applications. This arrangement illustrates how organization-wide coordination can coexist with the contextual judgment required at the application level. Importantly, this framework has not been evaluated for effectiveness, and the term “adaptive governance” refers to organizational adjustment rather than to system learning after deployment.

3.2 Ethical and Societal Ramifications

Governance misfit has ethical consequences when oversight loses its capacity to protect human stakeholders or provide recourse (Robbins, 2024; Wieringa, 2020). Retraining may change which patients or job candidates receive attention despite stable aggregate accuracy. Greater autonomy may allow a denial or suspension to take effect before human review. Impact assessment research explains why these outcomes need to be examined within the organizational practices that produce them (Metcalf et al., 2021; Stahl et al., 2023). The relevant question is how a change in system behavior or authority alters the positions of the people affected.

For example, a widely used health management algorithm predicted healthcare spending as a proxy for need, understating the needs of Black patients relative to White patients with the same risk scores (Obermeyer et al., 2019). This case shows how a technically useful prediction target can distribute access to care inequitably by establishing a mismatch between a proxy and a substantive objective, rather than through normative drift or a complete action–learning loop. For adaptive systems, humans must verify whether successive versions preserve the intended relationship between evaluation criteria and the interests of affected groups.

In a recent editorial, Richter et al. (2026) identified a related concern in situations where people remain formally accountable without having the capacity for practical judgment or authority necessary to challenge a system. They distinguished this accountability drift from capability

erosion, where reduced practice weakens human judgment, and from legitimacy gaps, where decisions cannot be adequately explained, defended, or revised. In line with this concern, our assessment of governance and fit extends beyond whether a named reviewer is present to whether that person can protect the interests of those affected.

What our framework of AI learning and action makes clear (particularly in relation to compound demands) is that recourse should be a key aspect of AI governance. As AI systems are becoming more autonomous, affected people need to be provided with practical ways to challenge AI-mediated decisions and correct them. A review of literature on operational AI revealed less attention to contestation and corrective review than to pre-deployment safeguards (Pang & Szliter, 2026). As a concrete example in the context of procurement, a supplier excluded based on an AI assessment needs a way to correct the evidence and escalate the issue to someone who is authorized to reconsider the exclusion, with explainability and timely correction designed into the workflow (Alfrink et al., 2023). For multi-step agents, organizations must also specify which actions may proceed without approval and how their effects can be stopped or reversed where possible (Shavit et al., 2023; Zhu et al., 2026).

Finally, our two-dimensional framework and insights regarding compounding demands complement contextual risk assessment. We used established characteristics to explain why learning after deployment and delegated action call for different safeguards, and why those safeguards may become interdependent. Applied across a portfolio, our framework can help organizations identify which assurances need to be renewed, which actions need tighter bounds, and which shared dependencies warrant a coordinated response.

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